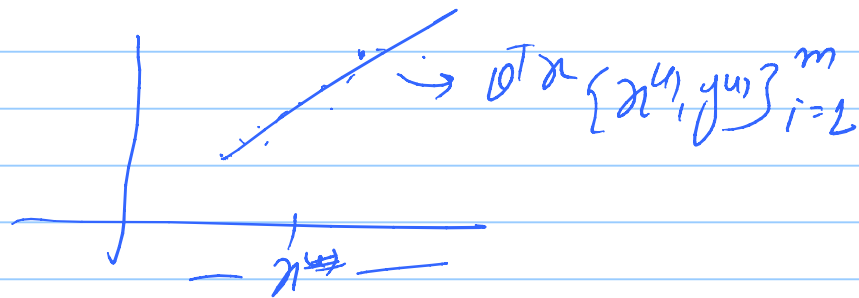


COL774 / 62734
Machine Learning
Aug 22, 2025

Probabilistic Interpretation of least squares (Regression)



$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (y^{(i)} - \theta^T x^{(i)})^2$$

$$y^{(i)} | x^{(i)}; \theta \sim \mathcal{N}(\theta^T x^{(i)}, \sigma^2)$$

$$h_{\theta}(x) = \theta^T x$$

↳ (unknown)

MLE
(Maximum Likelihood Estimation) $\Rightarrow p(y^{(1)} \dots y^{(m)} | x^{(1)} \dots x^{(m)}; \theta)$

$$L(\theta) = \frac{1}{n} \prod_{i=1}^m P(y^{(i)} | x^{(i)}; \theta)$$

$$L(\theta) = \frac{1}{n} \log \prod_{i=1}^m P(y^{(i)} | x^{(i)}; \theta)$$

$P(D; \theta)$ Likelihood of data

Logistic Regression:-

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

↳ indicator function

$\mathcal{H}$:- set of linear classifiers in $\mathbb{R}^n$

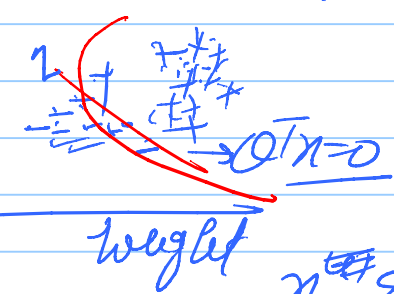
Equation of a hyperplane in $\mathbb{R}^n$ $\{ \theta^T x = 0 \}$

Classification:-

$$y \in \{1, -1\}$$

$$\hookrightarrow \{0, 1\}$$

Binary classification



Regression
 $y \in \mathbb{R}$

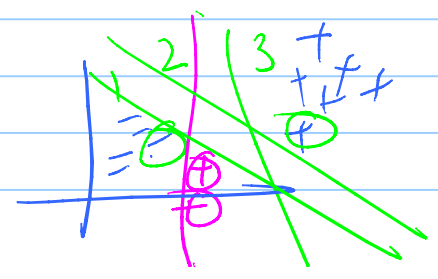
$$\{x^{(i)}, y^{(i)}\}_{i=1}^m$$

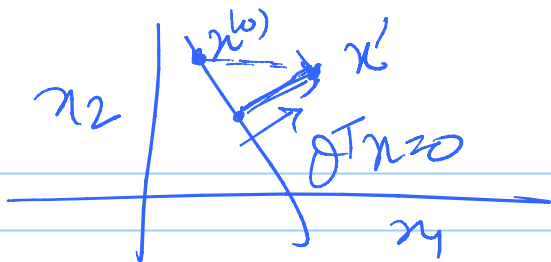
$$y \in \{1, -1\}$$

$$x \in \mathbb{R}^n$$

$$x = (x_0, \dots, x_n)$$

$$Q = (Q_0, \dots, Q_n)$$





$$\hat{d} = \theta^T x' \quad \boxed{d_0 + \bar{\theta}^T x^{(0)} = 0}$$

$\Downarrow$ unnormalized (signed) distance of x' from the line $\theta^T x = 0$

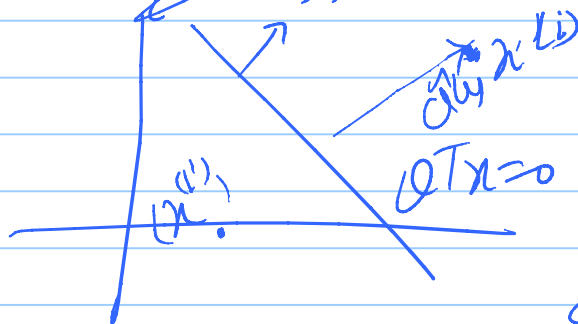
$$\bar{\theta} \equiv (\theta_1, \theta_2)$$

$$\theta \equiv (d_0, \bar{\theta})$$

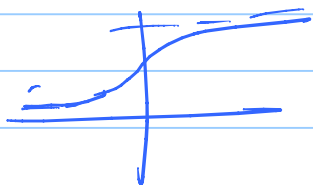
$$\begin{aligned} \bar{\theta}^T (x' - x^{(0)}) &= \bar{\theta}^T x' - \bar{\theta}^T x^{(0)} \equiv -d_0 \\ &= \bar{\theta}^T x' + d_0 \equiv \theta^T x' \Rightarrow \underline{\underline{\theta^T x^{(0)}}} \end{aligned}$$

$$y^{(i)} | x^{(i)}; \theta \sim \text{Bernoulli}(\phi^{(i)});$$

$$0 \leq \phi^{(i)} \leq 1$$



$$\text{the label } (x^{(i)}, y^{(i)}) \quad \{(x^{(i)}, y^{(i)})\}_{i=1}^m$$



$$\phi^{(i)} = \text{sigmoid}(\hat{d}^{(i)}) \rightarrow \theta^T x^{(i)}$$

